LLL-M-71b
December 19, 1963

SUPERSEDING
Fed. Spec. LLL-M-71a
September 4, 1956

FEDERAL SPECIFICATION

MALLETS AND MAULS, WOOD

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers wooden mallets and mauls (wood only).

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types and classes.* Mallets and mauls shall be of the following types and classes, as specified (see 6.2):

Type I.—Mallets, calking.
Type II.—Mallets, carpenters'.
  Class 1.—Round.
  Class 2.—Round, reinforced head.
  Class 3.—Square.
Type III.—Mallets, coppersmiths'.
Type IV.—Mallets, hawsing (beetles).
Type V.—Mallets, serving.
Type VI.—Mallets, tinners'.
Type VIII.—Mauls, ringbound.
Type IX.—Mauls, laminated-head.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-H-15424—Hand tools; Packaging of.
MIL-A-46051 — Adhesive, Room Temperature and Intermediate Temperature Setting Resin Phenol, Resorcinol and Melamine Base.

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Military Standard:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Illustrations.** The illustrations shown herein are descriptive and not restrictive and are included for the convenience of the requisitioning and purchasing officers and manufacturers, and are not intended to preclude the purchase of mallets and mauls which are otherwise in accordance with this specification.

3.2 **Material.** Material shall be as specified herein. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.3 **Wood.** The wood used in the fabrication of mallets and mauls shall be hardwood as specified herein. Moisture content shall be not more than 6 percent, except for type VIII mauls which shall have a moisture content of not more than 15 percent. The wood shall be sound and free from defects, except that up to four discolored lines or streaks not over 1/32 inch wide nor longer than one-third the length of the handle shall be acceptable in hickory handles; in handles of other species, mineral streak or other discoloration shall not be considered defects. Discolored lines or streaks associated with decay will not be permitted. Cross grain in the heads shall not exceed a slope of 1 to 8.

3.3.1 *Specific gravity (density).* When tested as specified in 4.4.5, the specific gravity of the wood shall be not less than 0.59 (37 pounds per cubic foot density).

3.4 **Tolerances.** Dimensions specified as "nominal" on the figures shall be interpreted as permitting a plus or minus tolerance of 10 percent from the dimensions specified.

3.5 **Handles.** Handles shall be well-proportioned and properly fitted, and shall be of conventional shape and design for the specified size, shape, class, and type of tool. Unless otherwise specified herein, the handle shall be secured in such a manner to preclude the probability of the head working loose or flying off the handle under severe use.

3.5.1 *Length.* The length of the handle shall be the overall length, including that portion which fits into the head.

3.6 **Handle holes.** Handle holes for all mallets and mauls shall be properly shaped, situated at right angles to the longitudinal axis of the head and to the grain of the wood, and shall be centrally located in the head to insure proper balance. Where handle hole sizes are not specified, the size and shape of the holes shall conform to the manufacturer's standard practice.

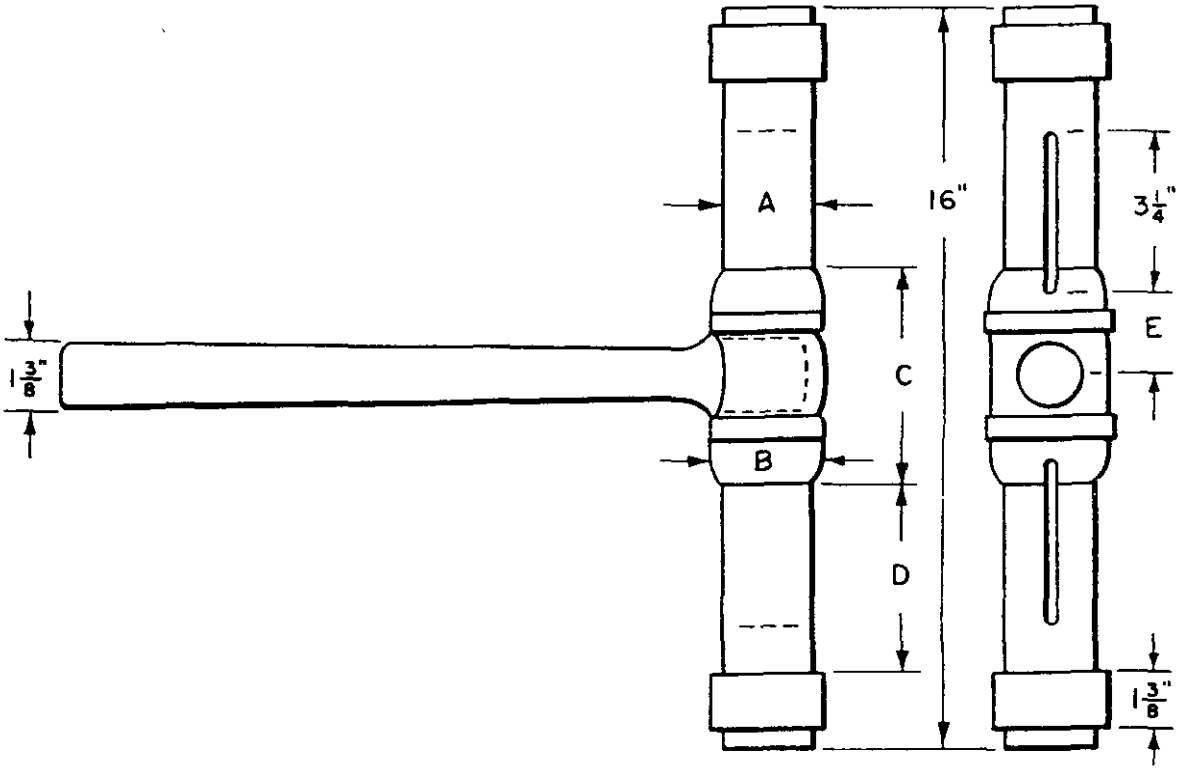
3.7 **Finish and protective treatment.** All handles and heads shall be finished smooth on all surfaces. (See 6.5 and Appendix A.)

3.8 **Performance.** Mallets and mauls shall not be damaged when subjected to the applicable tests in 4.4.

3.9 **Type I, calking mallets.** Calking mallets shall be similar to figure 1, and shall conform to the dimensions shown on the figure.

3.9.1 *Heads.* Heads shall be made of live oak, persimmon, applewood, dogwood, hard maple, black locust, rock elm, or black mesquite.

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NOMINAL DIMENSIONS

LENGTH OF HEAD	LENGTH OF HANDLE	DIAMETER OF HANDLE HOLE	A	B	C	D	E
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches
16	16	1 1/8	1 7/8	2 1/4	4 3/8	4 7/16	1 3/4

FIGURE 1.—Type I, calking mallets.

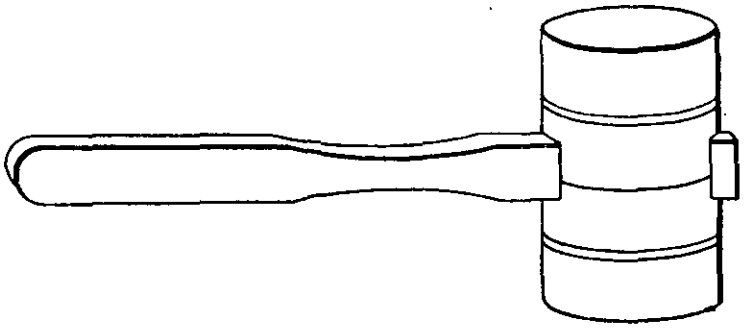
3.9.2 *Construction.* The heads of the mallets shall be reinforced at each end with a tool-steel ferrule having a nominal width of 1-3/8 inches and a wall thickness of not less than 1/16 inch, and a Rockwell hardness of between C40 and C50. The middle diameter of the head shall be reinforced with two soft steel bands, 1/2 inch wide and not less than 1/16-inch wall thickness. The soft steel bands shall be spaced not more than 2-1/4 inches, center to center, and spaced equidistant from the center of the handle hole. The bands and ferrules shall be firmly fastened to the mallet head. Each mallet shall be slotted longitudinally in two places, the slots extending through the axis from two sides of the mallet, and in the direction of the handle hole (see figure 1).

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3.9.3 *Handle.* The handles shall be made of hickory, hard maple, or tough white ash. The handle shall be accurately fitted to the hole in the head of the mallet, but shall not be secured. The handle shall extend beyond the head at least 1/4 inch when assembled. The slope of the grain shall be not more than one inch in 12 inches.

3.10 Type II, carpenters' mallets. Carpenters' mallets shall have heads made of hickory, dogwood, persimmon, applewood, lignum-vitae, rock elm, hard maple, hornbeam, live oak, locust, or black mesquite. Handles shall be made of hickory, tough white ash, or locust. The slope of the grain of the wood used in the handle shall be not more than one inch in 12 inches.

3.10.1 *Class* 1, *round.* Round carpenters' mallets shall be similar to figure 2, and shall conform to the dimensions shown on the figure.



NOMINAL DIMENSIONS

DIAMETER OF HEAD	LENGTH OF HEAD	LENGTH OF HANDLE
Inches	Inches	Inches
2½	4½	12
3	5	13
3½	5½	14
4	6	14

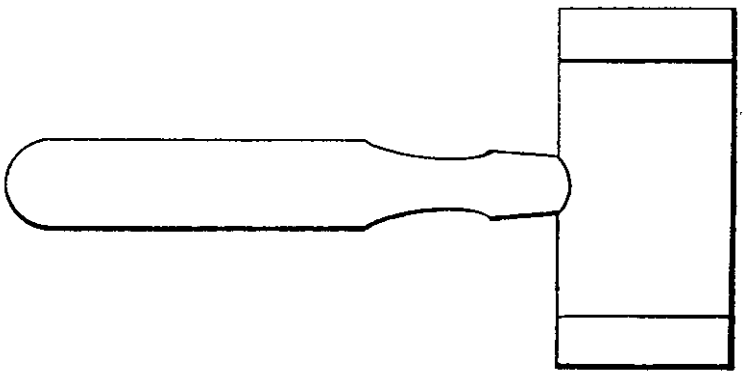
FIGURE 2.—Type II, class 1, round carpenters' mallets.

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3.10.1.1 *Construction.* The heads of the mallets shall be round and the faces parallel to each other. Handles shall be detachable, and shall be attached to the head with mortised or round tapered holes through the head. If round holes are used, the taper shall be between 1/2- and 3/4-inch per foot.

3.10.1.2 *Handle.* The handle shall be flattened on both sides at the handgrip.

3.10.2 *Class 2, round reinforced head.* Round reinforced-head mallets shall be similar to figure 3, and shall conform to the dimensions shown on the figure.



NOMINAL DIMENSIONS

DIAMETER OF HEAD	LENGTH OF HEAD	LENGTH OF HANDLE
Inches	Inches	Inches
4	6	14

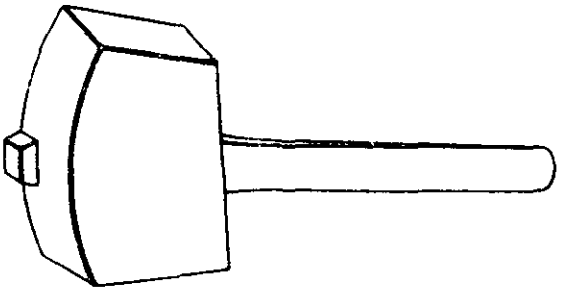
FIGURE 3.—Type II, class 2, round reinforced-head carpenters' mallets.

3.10.2.1 *Construction.* The heads of the mallets shall be round and the faces parallel to each other. Each end of the head shall be reinforced with a tool steel ferrule having a nominal width of one inch and a wall thickness of not less than 1/16 inch and a Rockwell hardness between C40 and C50. Handles need not be detachable.

3.10.2.2 *Handles.* The handles may be round in cross section or the sides may be flattened.

3.10.3 *Class 3, square.* Square carpenters' mallets shall be similar to figure 4, and shall conform to the dimensions shown on the figure.

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NOMINAL DIMENSIONS		
SIZE OF HEAD IN CROSS SECTION	LENGTH OF HEAD	LENGTH OF HANDLE
Inches	Inches	Inches
$2\frac{1}{2} \times 3\frac{1}{2}$	6	$10\frac{1}{2}$
$2\frac{3}{4} \times 3\frac{3}{4}$	$6\frac{1}{2}$	13
3 X 4	7	14

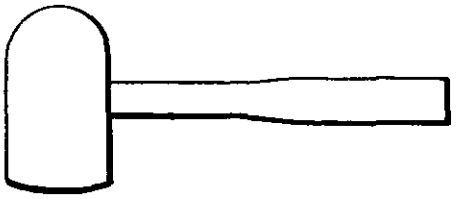
FIGURE 4.—Type II, class 3, square carpenters' mallets.

3.10.3.1 *Construction.* The heads shall be rectangular in cross section. Handles shall be detachable, and shall be attached to the head with mortised or round tapered holes through the head. If round holes are employed, the taper shall be between 1/2- and 3/4-inch per foot.

3.10.3.2 *Handles.* Handles may be round in cross section or flattened on the sides.

3.11 **Type III, coppersmiths' mallets.** Coppersmiths' mallets shall be similar to figure 5, and shall conform to the dimensions shown on the figure.

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NOMINAL DIMENSIONS

DIAMETER OF HEAD	LENGTH OF HEAD	LENGTH OF HANDLE
Inches	Inches	Inches
$2\frac{1}{4}$	$5\frac{1}{2}$	12
$3\frac{1}{2}$	7	16
5	10	24

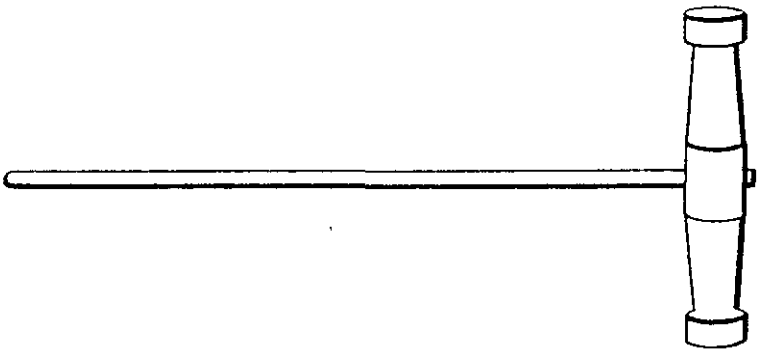
FIGURE 5.—Type III, coppersmiths' mallets.

3.11.1 *Heads.* The heads shall be made of hickory, lignum-vitae, black mesquite, dogwood, persimmon, applewood, locust, hard maple, rock elm, hornbeam or live oak. The heads may be either with or without bark. The head shall be round in cross section. One of the faces shall be slightly rounded, and the other face shall be semispherical.

3.11.2 *Handles.* The handles shall be round or oval in cross section, and shall be made from hickory, tough white ash, or locust. The slope of the grain shall be not more than one inch in 12 inches.

3.12 **Type IV, hawsing mallets (beetles).** Hawsing mallets shall be similar to figure 6, and shall conform to the dimensions shown on the figure.

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NOMINAL DIMENSIONS						
LENGTH OF HEAD	LENGTH OF BODY	LENGTH OF HANDLE	DIAMETER OF BODY	DIAMETER OF DRIVING FACES	LARGE DIAMETER OF HANDLE HOLE	WIDTH OF STEEL BANDS
Inches	Inches	Inches	Inches	Inches	Inches	Inches
16	4 3/4	32	3 3/8	2 7/8	1 1/4	1 1/4

FIGURE 6.—Type IV, hawsing mallets (beetles).

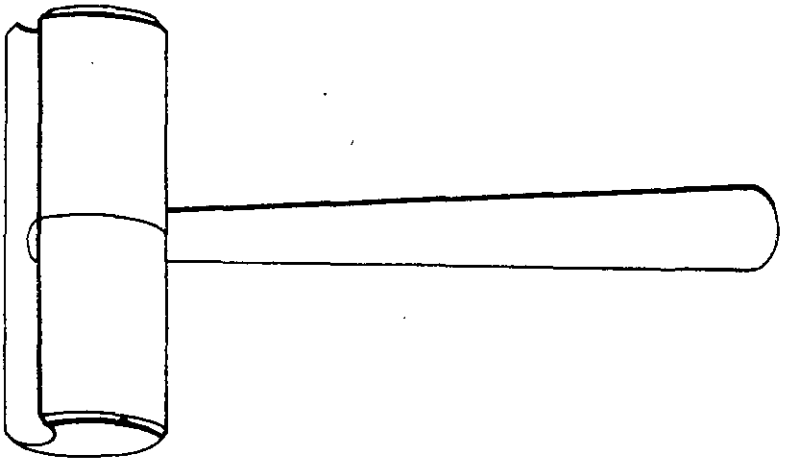
3.12.1 *Heads.* The heads shall be made of live oak.

3.12.2 *Construction.* The handle hole shall be double conical, with the smallest diameter at the center of the head, to permit reversing the handle. Two steel protective rivets of suitable size and strength shall be fitted into the body of the mallet diagonally opposite each other at an angle of about 45 degrees from the center of the handle hole. The rivets shall be so arranged as to give maximum support to the head of the mallet to prevent splitting of the wood when the mallet is in use under ordinary service conditions. The ends of the rivets shall be burred or riveted over washers. Steel protective bands of adequate strength shall be fitted snugly to the driving surfaces of the head. The outer edge of the steel band shall be flush 1/8 to 3/16 inch back from the driving surfaces of the mallet. The driving ends of the head shall taper slightly to insure tightness of the protective bands at all times as the driving surfaces wear away.

3.12.3 *Handles.* Handles shall be made of hickory. Handles shall be circular in cross section, and the mallet end shall be tapered to fit the head of the mallet. This slope of the grain shall be not more than one inch in 15 inches.

3.13 **Type V, serving mallets.** Serving mallets shall be similar to figure 7, and shall conform to the dimensions shown on the figure.

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NOMINAL DIMENSIONS

DIAMETER OF FACE	LENGTH OF MALLET	RADIUS OF GROOVE	DEPTH OF GROOVE	DIAMETER OF HANDLE AT LARGE END	LENGTH OF HANDLE	DIAMETER OF HANDLE HOLE
Inches	Inches	Inches	Inches	Inches	Inches	Inches
2	7	1/2	3/8	1 1/8	14	7/8
2 1/2	7 1/2	1/2	3/8	1 1/8	14	7/8
3	7 1/2	5/8	1/2	1 1/8	15	1

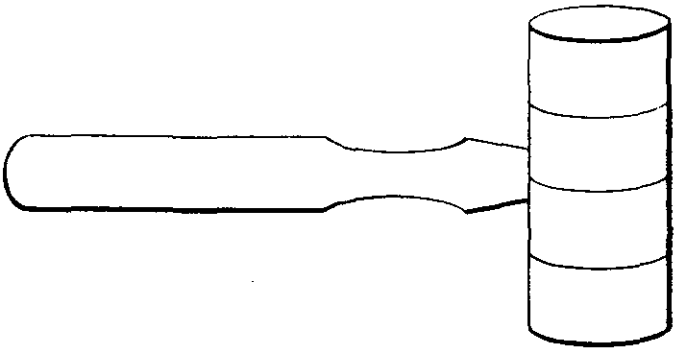
FIGURE 7.—Type V, serving mallets.

3.13.1 *Heads.* The heads shall be made of hickory, dogwood, persimmon, applewood, lignum-vitae, locust, hard maple, black mesquite, live oak, or rock elm. The heads shall be circular in shape and shall have a semicircular groove along the full length. The groove shall be located on the side of the mallet opposite the handle. The groove and the working surfaces shall be free from sharp edges and knots.

3.13.2 *Handles.* Handles shall be made of hickory, tough white ash, or locust. The handles shall be properly designed and firmly secured so that the end will not protrude into the groove of the head. The slope of the grain shall be not more than one inch in 12 inches.

3.14 Type VI, tinners' mallets. Tinners' mallets shall be similar to figure 8, and shall conform to the dimensions shown on the figure.

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NOMINAL DIMENSIONS

DIAMETER OF HEAD	LENGTH OF HEAD	LENGTH OF HANDLE
Inches	Inches	Inches
$2\frac{1}{4}$	$5\frac{1}{4}$	10
$2\frac{1}{2}$	$5\frac{1}{2}$	12
$2\frac{3}{4}$	$5\frac{3}{4}$	12
3	6	12
$3\frac{1}{2}$	6	13

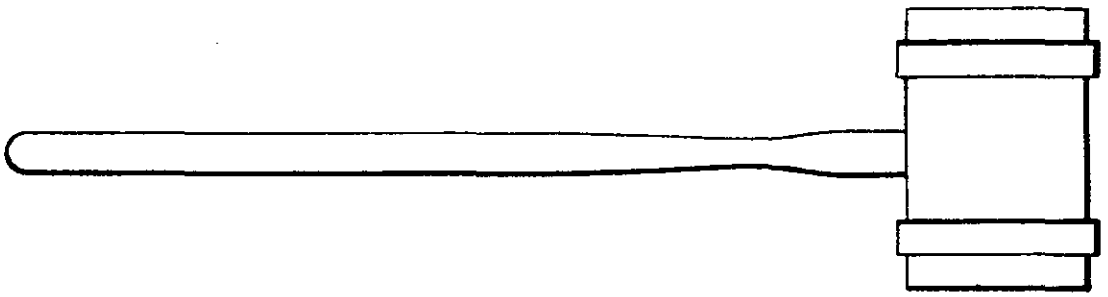
FIGURE 8.—Type VI, tinners' mallets.

3.14.1 *Heads.* Heads shall be made of dogwood, persimmon, hickory, applewood, locust, rock elm, hard maple, black mesquite, or live oak. Heads shall be either round or barrel-shaped with the faces parallel to each other and with the edges slightly rounded or beveled.

3.14.2 *Handles.* Handles shall be made of hickory, tough white ash, or locust. The slope of the grain shall be not more than one inch in 12 inches.

3.15 Type VIII, ringbound mauls. Ringbound mauls shall be similar to figure 9, and shall conform to the dimensions shown on the figure.

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NOMINAL DIMENSIONS

DIAMETER OF HEAD	LENGTH OF HEAD	LENGTH OF HANDLE
Inches	Inches	Inches
4	6	35
5	10	36
6	8	36
8	10	36

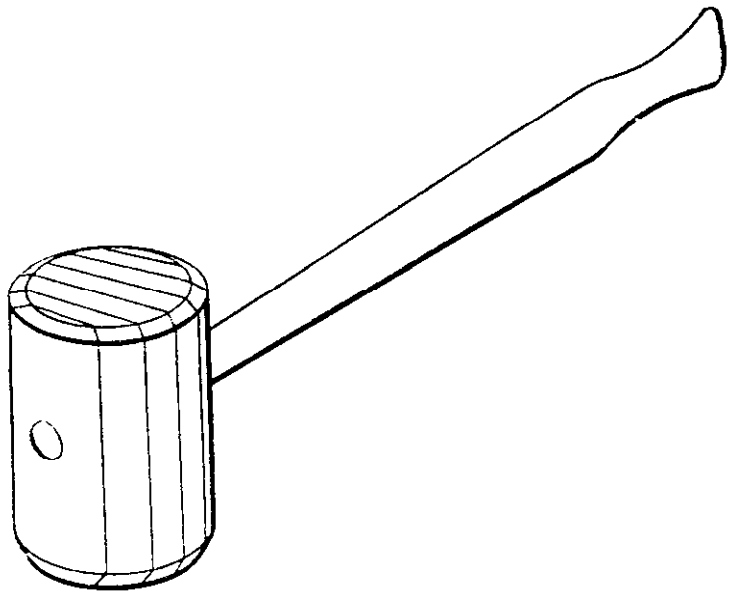
FIGURE 9 —Type VIII, ringbound mauls.

3.15.1 *Heads.* The heads shall be made of hickory, hard maple, persimmon, applewood, rock elm, dogwood, locust, black mesquite, or live oak. The heads shall be round and the faces shall be parallel to each other. Steel protective bands of adequate strength shall be shrunk in place near the driving face on each end of the head to give support to the head. The handle eye shall be elliptical shaped, 1-1/4 inches over its major axis and 1-1/16 inches over its minor axis.

3.15.2 *Handles.* Handles shall be made of sound, clear hickory. The slope of the grain shall be not more than one inch in 15 inches. The handle shall be elliptical shaped. The head end shall have cross section of 1-1/4 inches over its major axis and 1-1/16 inches over its minor axis for a distance of not less than 8 inches.

3.16 Type IX, laminated-head mauls. Laminated-head mauls shall be similar to figure 10, and shall conform to the dimensions shown on the figure.

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NOMINAL DIMENSIONS

DIAMETER OF HEAD	LENGTH OF HEAD	LENGTH OF HANDLE	DIAMETER OF HANDLE HOLE
Inches	Inches	Inches	Inches
6	8	36	$1\frac{3}{8}$
8	10	36	$1\frac{1}{2}$

FIGURE 10.—Type IX, laminated-head mauls.

3.16.1 *Heads.* The heads shall be made of kiln-dried maple, birch, beech, or white oak. When specified (see 6.2), steel protective bands shall be provided near the driving face of each end of the head to give support to the head. The handle eye shall be elliptical shaped, 1-1/4 inch over its major axis and 1-1/16 inch over its minor axis.

3.16.2 *Construction.* The heads shall be constructed of approximately 11 piles of hardwood glued together under adequate pressure and curing time to produce a serviceable maul. Each lamina shall be made up of solid pieces, or strips not less than 2-1/2 inches wide with abutting edges jointed and glued together before gluing all the lamina together to form the head. If a single lamina consists of two or more strips, the joints in the adjacent lamina shall be staggered. The grain of each lamina shall be parallel to the axis of the head. The head shall be cylindrical, with the faces parallel to each other, and with the edges slightly rounded. The handle shall be elliptical shaped. The head end shall have cross section of 1-1/4 inch over its major axis and 1-1/16 inch over its minor axis for a distance of not less than 8 inches.

3.16.2.1 *Adhesive.* Adhesive shall conform to MIL-A-46051.

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3.16.3 *Handles.* Handles shall be made of sound, clear hickory. The slope of the grain shall be not more than one inch in 15 inches. The handles shall pass completely through the head so that they will be perpendicular to the plane of the lamina and shall be firmly secured. The handle shall be removable for storage and replacement purposes.

3.17 **Marking.** The handles shall be marked in a plain and permanent manner, preferably foil or steel stamped, with the manufacturer's name, symbol or trademark of such known character that the source of manufacture may be readily determined.

3.18 **Workmanship.** Workmanship shall be in accordance with good commercial practice for this type of equipment.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling for quality conformance inspection.** Sampling for quality conformance inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.3 **End product inspection.**

4.3.1 *Visual examination.* Examination of the mallets and mauls shall be expressed in units of mallets or mauls for the purpose of determining the sample size in accordance with MIL-STD-105. The sample unit shall be one completely fabricated mallet or maul. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 defects per hundred units.

TABLE I.—*Classification of defects*

Examine	Defect
Material	Wood specie not as specified
Design	Not type and class as specified
Finish	Not smooth on all surfaces
	Evidence of checks, splits, wormholes, rot, dote or incipient decay, heart shake, torn grain or ragged end
	Pith of tree (if evident) not sound or diameter of pith exceeds the allowable maximum
	Evidence of unsound knot, diameter when sound, exceeds allowable maximum or located at working faces (head only)
	Angle of cross grain (or slope of grain) exceeds the allowable maximum as specified for head or handle
	Evidence of discolored lines or streaks (in hickory handles only) in excess of allowable maximum width and length
Construction and workmanship	Not accurately fitted to head, not detachable or secure in head, as applicable
	Head nonconforming; not the specified cross section
	Working faces of head not parallel to each other (type II, classes 1 and 2; types VI, VIII and IX only)
	Working faces of head not slightly rounded at one end and semispherical at other end (type III only)
	Bands or ferrules not firmly fastened
	Head not grooved, groove not opposite handle, groove does not extend the full length of the head, end of handle protrudes into groove or groove and working faces not free from sharp edges (type V only)
	Edges of working faces not rounded or beveled as specified (type IX only)

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TABLE I.—*Classification of defects (cont'd)*

Examine	Defect
Construction and workmanship (cont'd)	Laminations separated, number or width of laminations less than required, grain of laminations not parallel to axis of head, joints in adjacent strip when applicable not located as required, edges of working faces not slightly rounded, handle does not pass completely through the head (type IX only)
	Handle hole not at right angles with longitudinal axis of head and grain of wood, hole not centrally located

4.3.2 *Dimensional examination.* Inspection shall be made for compliance with dimensions specified in applicable paragraphs and tables. Any deviation therefrom shall constitute a defect. The inspection level shall be level II, with an AQL of 2.5 defects per hundred units.

4.3.3 *Testing of end item.* Testing shall be performed on each sample unit for characteristics specified in 4.4.2 through 4.4.4, as applicable. The inspection level shall be level L-6 with an AQL of 1.5 defects per hundred units.

4.3.4 *Testing of components and materials.* Tests shown in table II shall be performed to determine compliance with requirements specified in this specification.

TABLE II.—*Testing of components and materials*

Component	Characteristic	Test method	Sample size	Acceptance number
Handles and heads	Moisture content (prior to finishing)	4.4.1	10 from each lot or batch	0
Lumber (see 3.3.1)	Specific gravity	4.4.5	3 from each lot or batch	0

4.3.5 *Examination of preparation for delivery.* Inspection to determine compliance with preparation for delivery requirements shall be made in accordance with MIL-H-15424.

4.4 Test procedures.

4.4.1 *Moisture content.* The moisture content shall be determined by a standard moisture meter on both the handle and the head. The electrodes shall be driven to the center of the handle or head and a reading taken. Moisture meters shall be properly calibrated at all times. Nails may be used for electrodes when the readings are made on heads or handles of large dimensions.

4.4.2 *Transverse strength test.* The handle of each sample mallet or maul shall be required to pass a transverse strength test of at least 300 pounds without failure. The mallet or maul shall be laid on its side and the handle supported in a horizontal position while the 300-pound load is applied about two-thirds the distance from the outer end of the handle to the head of the mallet or maul. The load shall be applied normally at right angles to the longitudinal axis of the handle. In the absence of more suitable equipment an ordinary platform scale and pry bar may be employed in this test.

4.4.3 *Security of handle test.* Each sample type II, class 2, carpenters' mallet (having round handle hole), type III coppersmiths' mallet, type V serving mallet, and type VI tinners' mallet, shall also be subjected to the following test: Place the handle in a verticle position with the handle resting on a circular object slightly smaller in diameter than the handle hole in the mallet or maul. The circular object shall rest on a solid foundation and shall have a height of not less than 3/4 inch. With the mallet or maul in this position use a circular weight to fit over the handle, having a nominal weight of 4 pounds. The weight shall be dropped at least six times from the top

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or outer end of the handle so that it will impinge upon the head in a manner that will tend to force it off the handle.

4.4.4 *Tightness of ringbands and ferrules.* Mallets and mauls equipped with ringbands or ferrules shall be subjected to a practical test of sufficient duration to satisfy the inspector as to whether the bands and ferrules fit the head securely.

4.4.5 *Specific gravity.* Specific gravity shall be determined by ascertaining the ratio of the volume below the waterline to that of the whole piece. The determination of specific gravity by this method is done as follows:

(*a*) Cut a piece of wood from the lumber to be tested, about 1- by 1-inch square by 12 inches long, the long dimension being along the grain.

(*b*) Dry the piece for 48 hours at 212° to 221°F. heat.

(*c*) Dress the piece so that the cross section is rectangular and uniform throughout its length. Cut the piece to exactly 10 inches in length. Mark and number the length in 1-inch demarcations.

(*d*) Place the piece carefully in a tall glass container of water so that it floats in an upright position. Note the water line to which it sinks. Quickly remove and mark the average wetted length. The position of the water line indicates the specific gravity of the piece on the basis of oven-dry weight and oven-dry volume. Thus, if the mark is halfway between the 5- and 6-inch mark the specific gravity is 0.55. The whole operation must be performed quickly, because the oven-dry piece will begin to pick up moisture while it is being dressed, cut, marked, and when it is placed in the water. When determining the acceptability of suspected material the required immersion depth may be marked on the specimen before testing.

5. PREPARATION FOR DELIVERY

(Civil agencies should refer to Fed. Std. No. 102 for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Cleaning, preservation, and packaging (see 6.2).

5.1.1 *Level A.* Mallets and mauls shall be preserved and packaged in accordance with level A of MIL-H-15424.

5.1.2 *Level C.* Cleaning, preservation, and packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2 Packing. (See 6.2.)

5.2.1 *Level A.* Mallets and mauls, packaged as specified (see 6.2), shall be packed in accordance with level A requirements of MIL-H-15424.

5.2.2 *Level B.* Mallets and mauls, packaged as specified (see 6.2), shall be packed in accordance with level B requirements of MIL-H-15424.

5.2.3 *Level C.* Mallets and mauls, packaged as specified (see 6.2), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or other regulations as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Military activities.* In addition to any special marking required by the contract or order, or herein, interior and exterior shipping containers shall be marked in accordance with MIL-H-15424.

5.3.2 *Civil agencies.* In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. The mallets and mauls covered by this specification are intended

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for use by boatbuilders, carpenters, riggers, bridge erection crews, coppersmiths, and tinsmiths in striking chisels, marlinspikes, pins, and beating metals.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and size of mallet or maul required (1.2.1).
(*c*) Class required, type II mallets only.
(*d*) Whether bands shall be provided for type IX mauls (3.16.1).
(*e*) Applicable levels of preservation, packaging, and packing (5.1 and 5.2).
(*f*) Special marking, if required (5.3).

6.3 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Mallets:

Rail:

Mallets, wooden.
Carload minimum weight 30,000 pounds.

Motor:

Mallets, wooden.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Mauls:

Rail:

Mauls, handled.
Carload minimum weight 30,000 pounds.

Motor:

Mauls, handled.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.4 **Deletion.** Type VII, rawhide mallet, has been deleted from this specification and placed in Federal Specification GGG-H-33a.

6.5 **Treatment.** Sometimes rough bolts as well as unfinished squares and stored handles are severely damaged by infestations of insects, generally Lyctus powder-post beetles. Bolts cut during the season when insects are active are subject to attack by Lyctus beetles as soon as the moisture content is below 20 percent. This is also true of unfinished squares or manufactured but unfinished handle stock and similar items. Fortunately Lyctus attack can be prevented by dipping the wood items in an insecticide. To be effective, the insecticide must be applied to raw wood since it will not penetrate paints, varnishes, lacquers, waxes, or other finishing materials. For those who are interested in insecticide dips, a consideration of Appendix A is suggested. The methods described therein may be specified in connection with the procurement of mallets and mauls or they may be followed by the purchaser. Lyctus infestations can be very costly. Adequate treatments are relatively inexpensive and should be included as part of the manufacturing process.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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APPENDIX A. — TREATMENT TO PREVENT INFESTATION

1. Treatment.

1.1 **Turned handles.** After turning and before installing ferrules or metal rings, as applicable, dip the heads and handles for 3 minutes in a solution of DDT, conforming to O-I-531. Care shall be exercised to prevent DDT solution from coming into contact with those dipping. The heads and handles shall be allowed to drain and dry before shipping or storing.

1.2 **Bolts and unmanufactured squares.** Green bolts may be protected from Lyctus infestation by giving them a 10-second dip in a 5 percent water emulsion of DDT. Seasoned (10 to 15 percent moisture content) bolts and unmanufactured squares may be treated as stipulated for turned handles and heads.

2. **Determination of the presence of DDT.** From a treated handle or head, plane off 1/4 inch and discard the shavings. Take the next fairly thick shaving for submission to a selected laboratory for testing as follows:

2.1 **Preparation of reagent.** Seventy-five milliliters (ml.) of reagent grade pyridine and 50 grams of potassium hydroxide pellets shall be heated for 1 hour on a steam bath in a liter Erlenmeyer flask, fitted with an all glass reflux condenser. The pyridine shall be cooled and decanted into a second Erlenmeyer flask containing an additional 50 grams of potassium hydroxide and shall then be cooled, decanted and distilled in an all glass distillation apparatus. Two-tenths percent water and 0.2 percent xanthydrol shall be added to the distillate.

2.2 **Test procedure.** Scrapings or thin shavings from the handle or head shall be placed in a beaker and covered with benzene. The benzene shall be brought to a boil and then filtered into a test tube. The benzene solution shall be evaporated to dryness. Five ml. of the pyridine zanthydrol solution and four pellets of potassium hydroxide shall be added and heated to boiling for 3 to 5 minutes. The reagent alone develops a yellow color changing to a dark green which on cooling reverts to a light greenish yellow color. When DDT is present, a red color develops first, followed by a dark green.

Copies of this specification may be purchased for 15 cents each.

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